



# Chapter 9: NAT for IPv4



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# NAT Operation

# IPv4 Private Address Space

- Private IP addresses are used within an organization's internal networks.

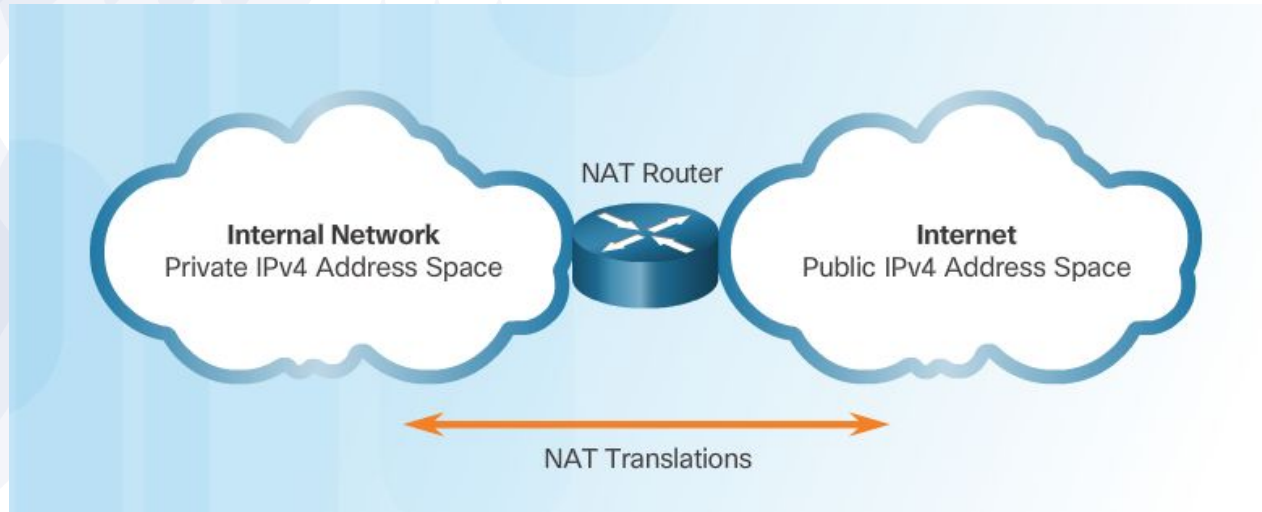
Did you ever notice  
how all your labs were  
based on these  
addresses?

### Private Internet Addresses are Defined in RFC 1918

| Class | RFC 1918 Internal Address Range | CIDR Prefix    |
|-------|---------------------------------|----------------|
| A     | 10.0.0.0 - 10.255.255.255       | 10.0.0.0/8     |
| B     | 172.16.0.0 - 172.31.255.255     | 172.16.0.0/12  |
| C     | 192.168.0.0 - 192.168.255.255   | 192.168.0.0/16 |

These are the IP addresses you will see assigned to company devices.

- Private IP addresses cannot be routed over the Internet.
- NAT is used to translate private IP addresses to public addresses that can be routed over the Internet.
- One public IPv4 address can be used for thousands of devices that have private IP addresses.



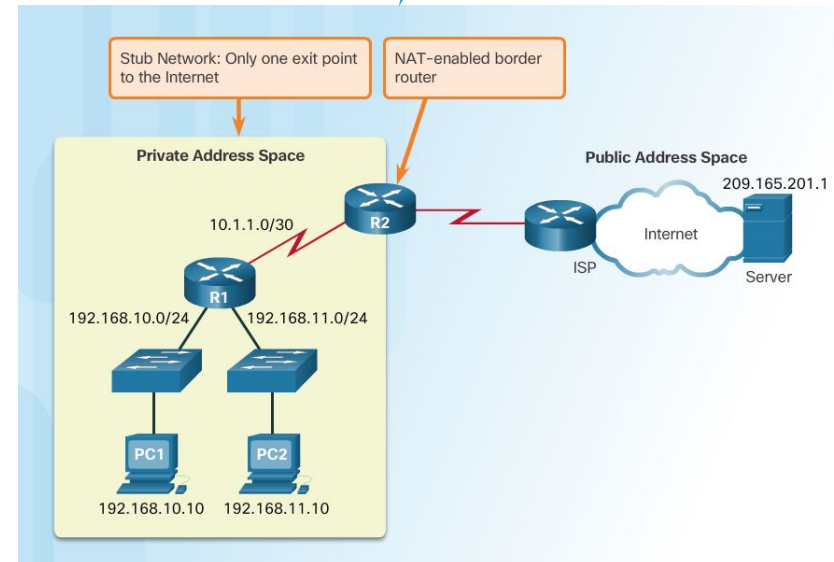
# NAT Characteristics

## What is NAT?

- Private IP addresses cannot be routed over the Internet.
- NAT is used to translate private IP addresses used inside a company to public addresses that can be routed over the Internet.
- NAT hides internal IPv4 addresses from outside networks.
  - Companies use the same private IPv4 addresses so outside devices cannot tell one company's 10.x.x.x network from another company's 10.x.x.x network.
- A NAT-enabled router can be configured with a public IPv4 address.
- A NAT-enabled router can be configured with multiple public IPv4 addresses to be used in a pool or NAT pool for internal devices configured with private addresses.



Important Concept—NAT is enabled on one device (normally the border or edge router)



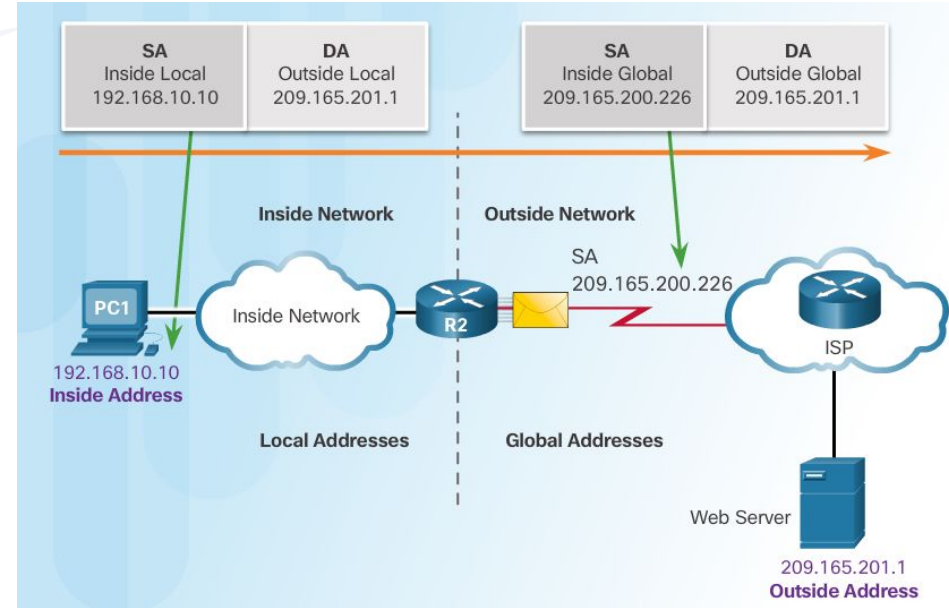
# NAT Characteristics

## NAT Terminology



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- Four types of addresses:
  - inside, outside, local, and global
  - Always consider the device that is having its private address translated to understand this concept.
  - Inside address – address of the company network device that is being translated by NAT
  - Outside address – IP address of the destination device
  - Local address – any address that appears on the inside portion of the network
  - Global address – any address that appears on the outside portion of the network

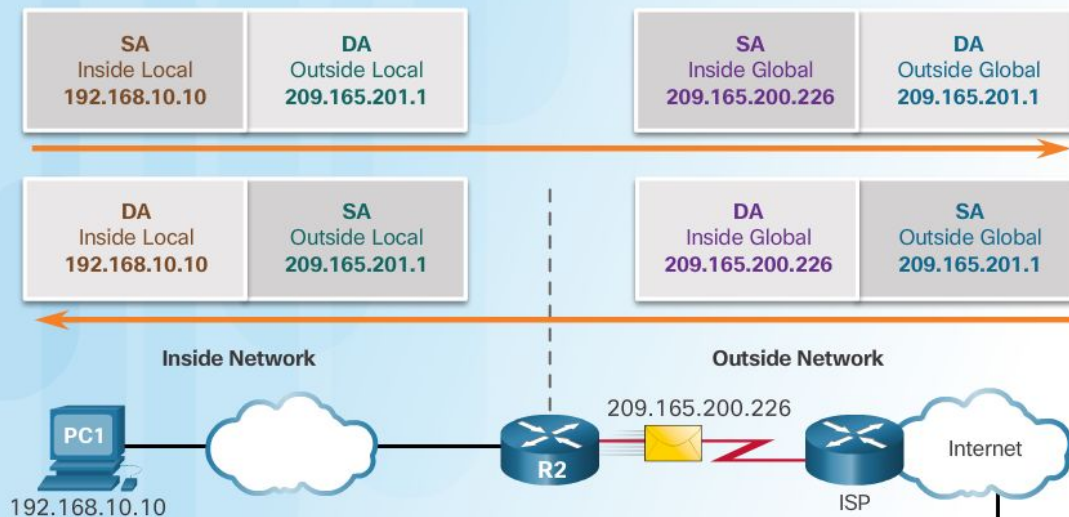


# NAT Characteristics

## NAT Terminology (Cont.)



### NAT Address Examples



# NAT Characteristics

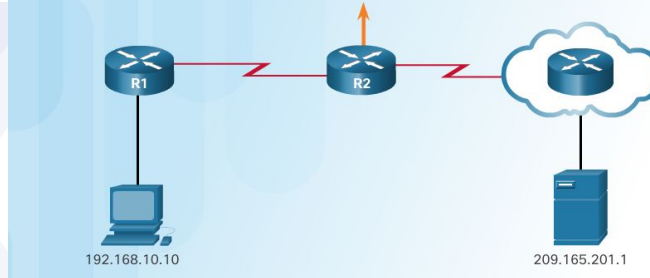
## How NAT Works



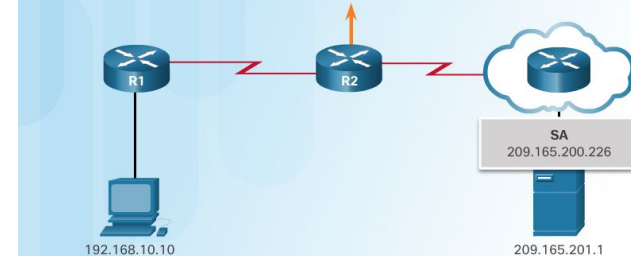
1. The private (internal) IP address gets translated to a public IP address used to reach the external server.



| NAT Table     |                 |               |                |
|---------------|-----------------|---------------|----------------|
| Inside Local  | Inside Global   | Outside Local | Outside Global |
| 192.168.10.10 | 209.165.200.226 | 209.165.201.1 | 209.165.201.1  |



| NAT Table     |                 |               |                |
|---------------|-----------------|---------------|----------------|
| Inside Local  | Inside Global   | Outside Local | Outside Global |
| 192.168.10.10 | 209.165.200.226 | 209.165.201.1 | 209.165.201.1  |



# NAT Characteristics

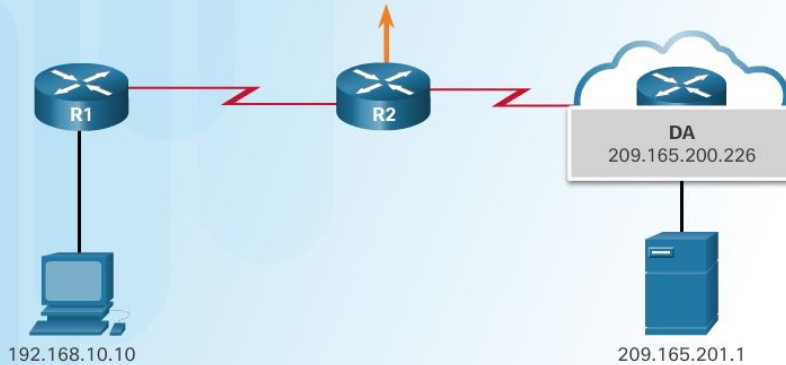
## How NAT Works (Cont.)



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**NAT Table**

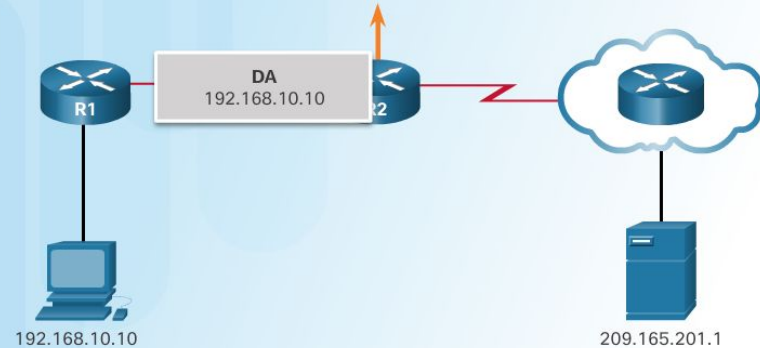
| Inside Local  | Inside Global   | Outside Local | Outside Global |
|---------------|-----------------|---------------|----------------|
| 192.168.10.10 | 209.165.200.226 | 209.165.201.1 | 209.165.201.1  |



2. The translated public address is used by the server to send the requested information to the device that actually has a private IP address assigned to it.

**NAT Table**

| Inside Local  | Inside Global   | Outside Local | Outside Global |
|---------------|-----------------|---------------|----------------|
| 192.168.10.10 | 209.165.200.226 | 209.165.201.1 | 209.165.201.1  |



3. The NAT-enabled router consults the routing table to see what private address requested the data.

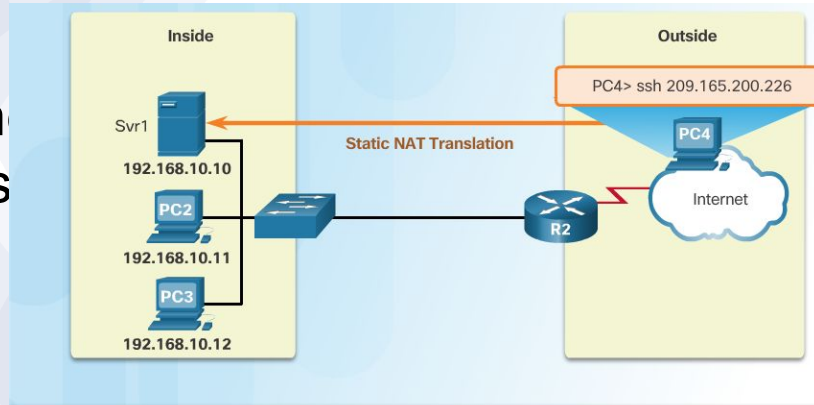
# Types of NAT

## Static NAT



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- Static address translation (static NAT) assigns one public IP address to one private IP address
- Commonly used for servers that need to be accessed by external devices or for devices that must be accessible by authorized personnel
- One-to-one addresses



and global

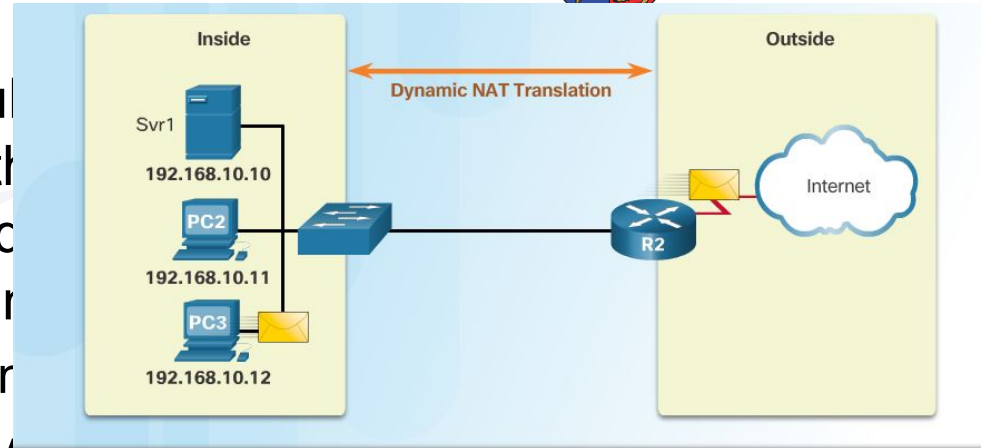
Static NAT Table

| Inside Local Address | Inside Global Address - Addresses reachable via R2 |
|----------------------|----------------------------------------------------|
| 192.168.10.10        | 209.165.200.226                                    |
| 192.168.10.11        | 209.165.200.227                                    |
| 192.168.10.12        | 209.165.200.228                                    |

## Types of NAT

### Dynamic NAT

- Dynamic NAT assigns a public address to each packet that has a private IP address assigned to it. The network outside the company
  - Addresses are assigned on a first-come, first-served basis.
  - The number of internal devices that can access the Internet from the company is limited to the number of public addresses in the pool.



IPv4 NAT Pool

| Inside Local Address | Inside Global Address Pool - Addresses reachable via R2 |
|----------------------|---------------------------------------------------------|
| 192.168.10.12        | 209.165.200.226                                         |
| Available            | 209.165.200.227                                         |
| Available            | 209.165.200.228                                         |
| Available            | 209.165.200.229                                         |
| Available            | 209.165.200.230                                         |

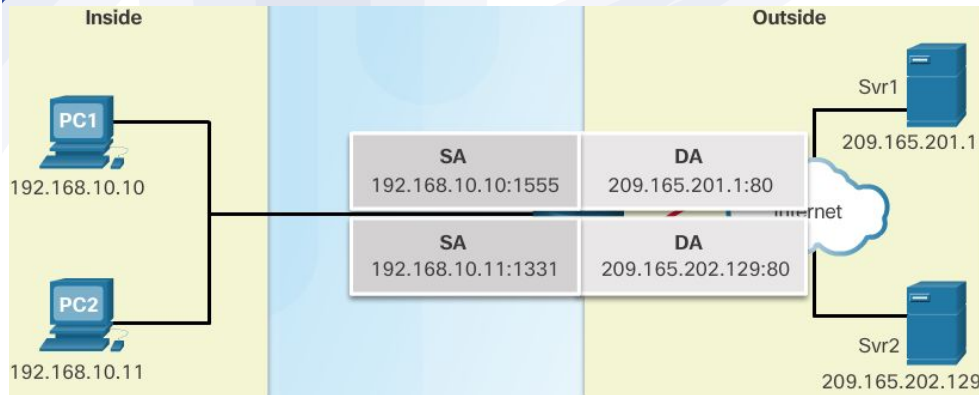
# Types of NAT

## Port Address Translation (PAT)

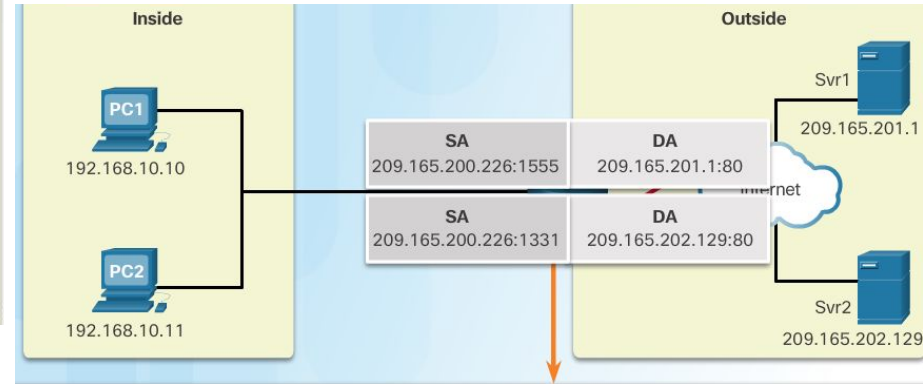


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- PAT (otherwise known as NAT overload) can use one public IPv4 address to allow thousand of private IPv4 addresses to communicate with outside network devices.



### Session



**NAT Table with Overload**

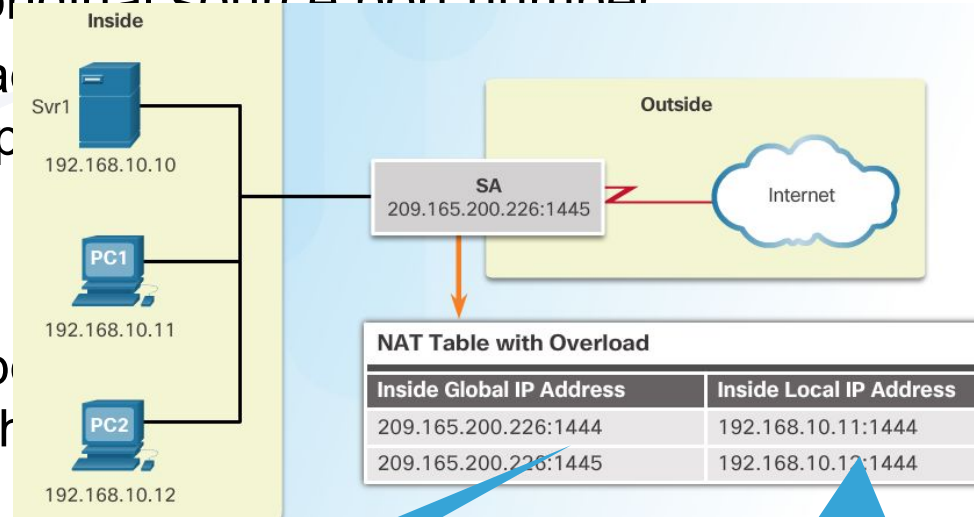
| Inside Global IP Address | Inside Local IP Address | Outside Local IP Address | Outside Global IP Address |
|--------------------------|-------------------------|--------------------------|---------------------------|
| 209.165.200.226:1555     | 192.168.10.10:1555      | 209.165.201.1:80         | 209.165.201.1:80          |
| 209.165.200.226:1331     | 192.168.10.11:1331      | 209.165.202.129:80       | 209.165.202.129:80        |

## Types of NAT

### Next Available Port



- PAT tries to preserve the original source port number
  - If that port number is already in use, PAT uses the next available port number
    - 0 - 511
    - 512 - 1023
    - 1024 - 65,535
  - When there are no more ports available, PAT uses the next public IP address in the pool



2. Notice how PAT uses the same public address, but two different port numbers.

1. Notice how traffic is from two different internal devices using the same port number.

# Types of NAT

## Comparing NAT and PAT



- Static NAT translates address on a 1:1 basis

### NAT

| Inside Global Address Pool | Inside Local Address |
|----------------------------|----------------------|
| 209.165.200.226            | 192.168.10.10        |
| 209.165.200.227            | 192.168.10.11        |
| 209.165.200.228            | 192.168.10.12        |
| 209.165.200.229            | 192.168.10.13        |

so that one public address can be used  
addressed devices

with a protocol such as ICMP that does not

### PAT

| Inside Global Address | Inside Local Address |
|-----------------------|----------------------|
| 209.165.200.226:1444  | 192.168.10.10:1444   |
| 209.165.200.226:1445  | 192.168.10.11:1444   |
| 209.165.200.226:1555  | 192.168.10.12:1555   |
| 209.165.200.226:1556  | 192.168.10.13:1555   |

# Advantages of NAT



- Conserves the legally registered addressing scheme
  - Every company can use the private IP addresses
- Increases the flexibility of connections to the public network
  - Multiple NAT pools, backup pools, and load-balancing across NAT pools
- Provides consistency for internal network addressing schemes
  - Do not have to readdress the network if a new ISP or public IP address is assigned
- Provides network security
  - Hides user private IPv4 addresses



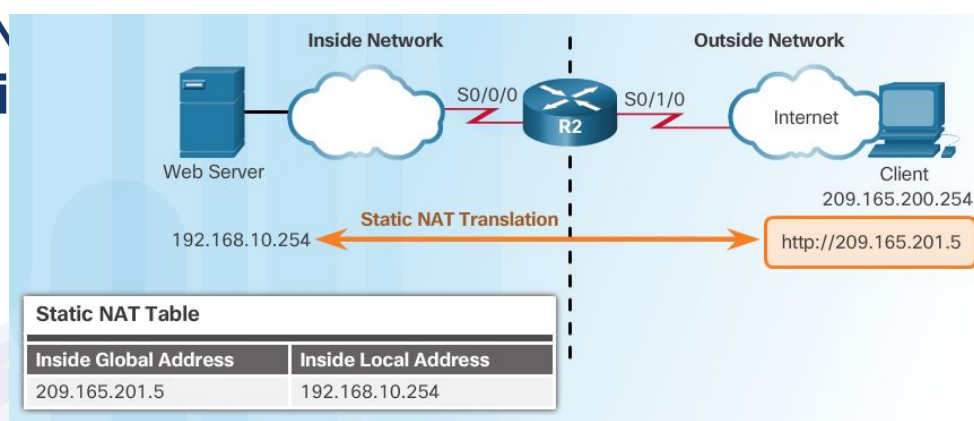
- Performance is degraded.
  - The NAT-enabled border device must track and process each session destined for an external network.
- End-to-end functionality is degraded.
  - Translation of each IPv4 address within the packet headers takes time.
- End-to-end IP traceability is lost.
  - Some applications require end-to-end addressing and cannot be used with NAT.
  - Static NAT mappings can sometimes be used.
  - Troubleshooting can be more challenging.
- Tunneling becomes more complicated.
- Initiating TCP connections can be disrupted.



# Configure NAT

# Configuring Static NAT

## Configure Static



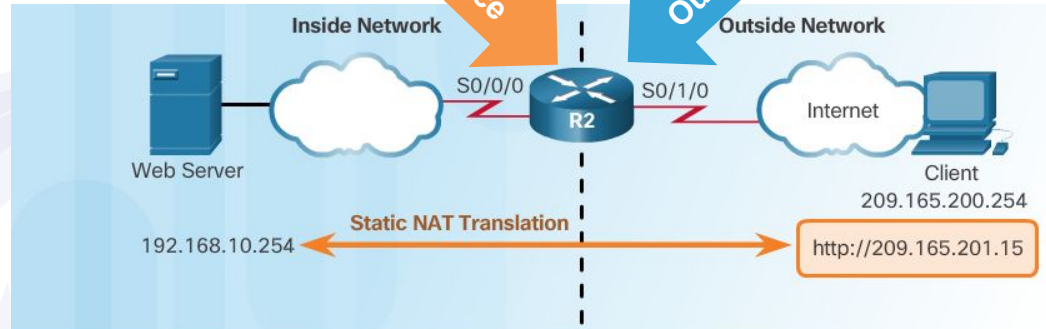
| Step | Action                                                                                                                                                              | Notes                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1    | Establish static translation between an inside local address and an inside global address.<br>Router(config)# <b>ip nat inside source static local-ip global-ip</b> | Enter the <b>no ip nat inside source static</b> global configuration mode command to remove the dynamic source translation. |
| 2    | Specify the inside interface.<br>Router(config)# <b>interface</b> type number                                                                                       | Enter the <b>interface</b> command. The CLI prompt changes from (config)# to (config-if)#.                                  |
| 3    | Mark the interface as connected to the inside.<br>Router(config-if)# <b>ip nat inside</b>                                                                           |                                                                                                                             |
| 4    | Exit interface configuration mode.<br>Router(config-if)# <b>exit</b>                                                                                                |                                                                                                                             |
| 5    | Specify the outside interface.<br>Router(config)# <b>interface</b> type number                                                                                      |                                                                                                                             |
| 6    | Mark the interface as connected to the outside.<br>Router(config-if)# <b>ip nat outside</b>                                                                         |                                                                                                                             |

# Configuring Static NAT

## Configure Static NAT (Con



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Remember that any interface on the border router that is on the inside network must be configured with the **ip nat inside** command. This is a common mistake for those new to NAT.

Establishes static translation between an inside local address and an inside global address.

```
R2(config)# ip nat inside source static 192.168.11.99 209.165.201.15
```

```
R2(config)# interface Serial0/0/0
```

```
R2(config-if)# ip address 10.1.1.2 255.255.255.252
```

Identifies interface serial 0/0/0 as an inside NAT interface.

```
R2(config-if)# ip nat inside
```

```
R2(config-if)# exit
```

```
R2(config)# interface Serial0/1/0
```

```
R2(config-if)# ip address 209.165.200.1 255.255.255.252
```

Identifies interface serial 0/1/0 as the outside NAT interface.

```
R2(config-if)# ip nat outside
```

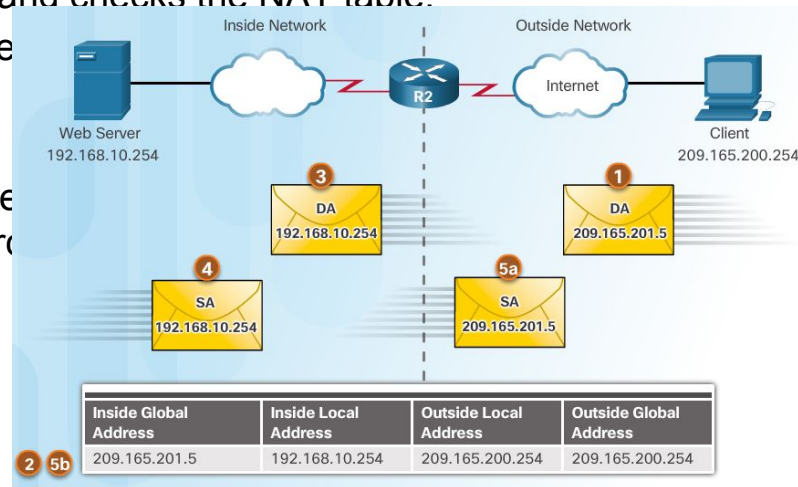
# Configuring Static NAT

## Analyzing Static NAT



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1. Client opens a web browser for a connection to a web server.
2. R2 receives the packet on the outside interface and checks the NAT table.
3. R2 replaces the inside global address with inside address).
4. Web server responds to the client.
5. (a) R2 receives the packet from the server on the  
(b) R2 checks NAT table and translates the source 209.165.201.5 and forwards the packet.
6. The client receives the packet.



# Configuring Static NAT

## Verifying Static NAT

A best practice is to clear statistics when verifying that NAT is working.

The static translation is always present in the NAT table.

```
R2# show ip nat translations
```

| Pro | Inside global | Inside local   | Outside local | Outside global |
|-----|---------------|----------------|---------------|----------------|
| --- | 209.165.201.5 | 192.168.10.254 | ---           | ---            |

```
R2#
```

The static translation during an active session.

```
R2# show ip nat translations
```

| Pro | Inside global | Inside local   | Outside local   | Outside global  |
|-----|---------------|----------------|-----------------|-----------------|
| --- | 209.165.201.5 | 192.168.10.254 | 209.165.200.254 | 209.165.200.254 |
| --- | 209.165.201.5 | 192.168.10.254 | ---             | ---             |

```
R2#
```

Important commands:

- **show ip nat translations**
- **show ip nat statistics**

```
R2# clear ip nat statistics
```

```
R2# show ip nat statistics
```

```
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/0/0
Hits: 0 Misses: 0
<output omitted>
```

Client PC establishes a session with the web server

```
R2# show ip nat statistics
```

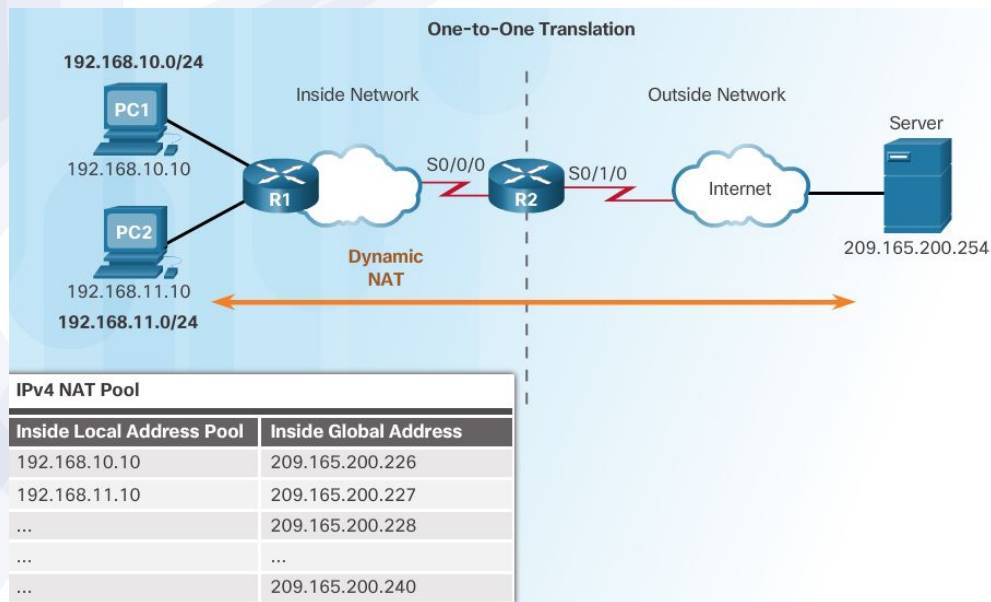
```
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
Peak translations: 2, occurred 00:00:14 ago
Outside interfaces:
  Serial0/1/0
Inside interfaces:
  Serial0/0/0
Hits: 5 Misses: 0
<output omitted>
```

# Configure Dynamic NAT

## Dynamic NAT Operation



- Remember that dynamic NAT uses a pool of public IPv4 addresses.
- Use the same concepts of inside and outside NAT interfaces as static NAT.



# Configure Dynamic NAT

## Configuring Dynamic NAT



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### Dynamic NAT Configuration Steps

**Step 1**

Define a pool of global addresses to be used for translation.

```
ip nat pool  name start-ip end-ip  
{netmask netmask | prefix-length  prefix-length}
```

**Step 2**

Configure a standard access list permitting the addresses that should be translated.

```
access-list  access-list-number  permit  source [source-wildcard]
```

**Step 3**

Establish dynamic source translation, specifying the access list and pool defined in prior steps.

```
ip nat inside source list  access-list-number  pool  name
```

**Step 4**

Identify the inside interface.

```
interface  type number  
ip nat inside
```

**Step 5**

Identify the outside interface.

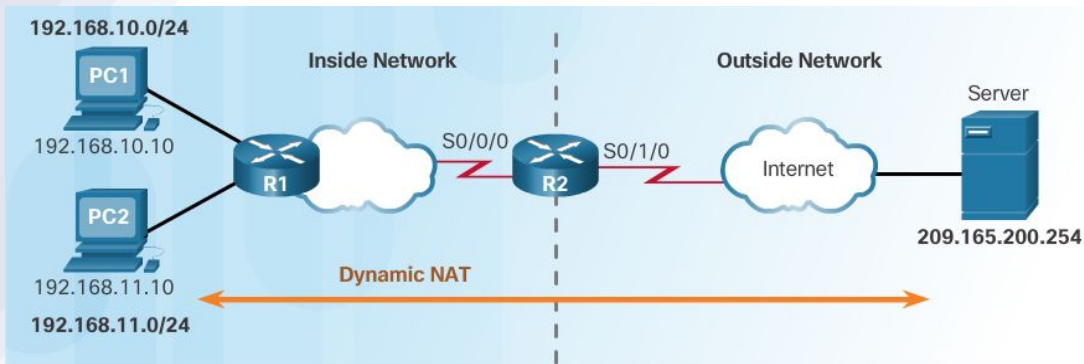
```
interface  type number  
ip nat outside
```

# Configure Dynamic NAT

## Configuring Dynamic NAT (Cont.)



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Defines a pool of public IPv4 addresses under the pool name NAT-POOL1.

```
R2(config)# ip nat pool NAT-POOL1 209.165.200.226  
209.165.200.240 netmask 255.255.255.224
```

Defines which addresses are eligible to be translated.

```
R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```

Binds NAT-POOL1 with ACL 1.

```
R2(config)# ip nat inside source list 1 pool NAT-POOL1
```

Identifies interface serial 0/0/0 as an inside NAT interface.

```
R2(config)# interface Serial0/0/0
```

```
R2(config-if)# ip nat inside
```

Identifies interface serial 0/1/0 as an outside NAT interface.

```
R2(config)# interface Serial0/1/0
```

```
R2(config-if)# ip nat outside
```

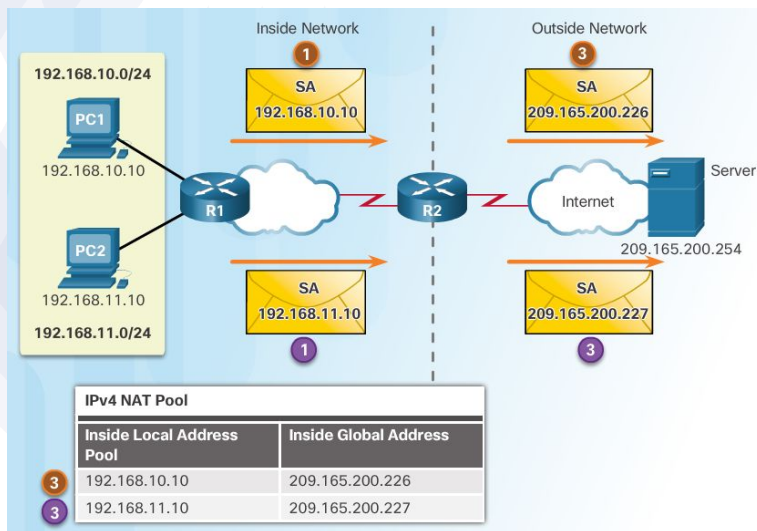
# Configure Dynamic NAT

## Analyzing Dynamic NAT



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1. PC1 and PC2 open a web browser for a connection to a web server.
2. R2 receives the packets on the inside interface and checks if translation should be performed (via an ACL). R2 assigns a global address from the NAT pool and creates a NAT table entry for both packets.
3. R2 replaces the inside local source address on each packet with the translated inside global address from the pool.



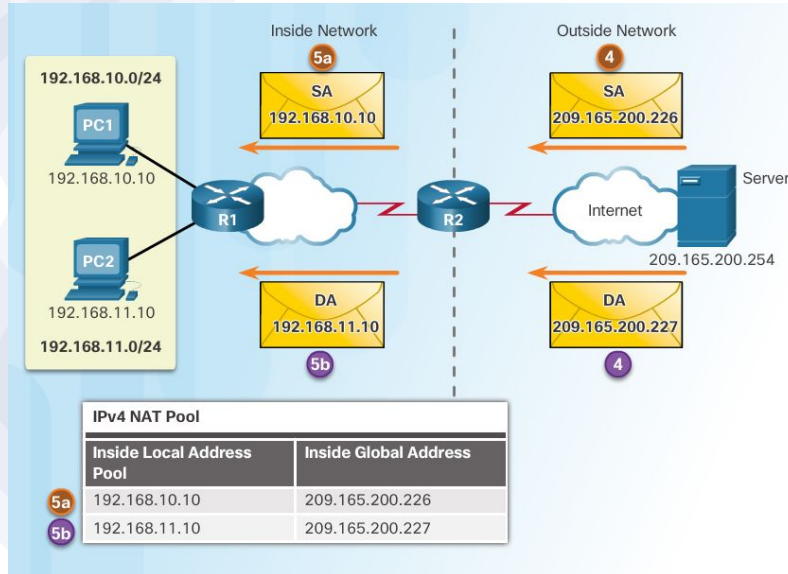
# Configure Dynamic NAT

## Analyzing Dynamic NAT (Cont.)



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4. The server responds to PC1 using the destination address of 209.165.200.226 (the NAT-assigned address) and to PC2 using the destination address of 209.165.200.227.
5. (a and b) R2 looks up each received packet and forwards based on the private IP address found in the NAT table for each of the destination addresses.



# Configure Dynamic NAT

## Verifying Dynamic NAT



```
R2# clear ip nat translation *
R2# show ip nat translations
```

```
R2# show ip nat translations
Pro Inside global    Inside local  Outside local  Outside global
--- 209.165.200.226  192.168.10.10 ---          ---
--- 209.165.200.227  192.168.11.10 ---          ---
R2#
R2# show ip nat translations verbose
Pro Inside global    Inside local  Outside local  Outside global
--- 209.165.200.226  192.168.10.10 ---          ---
    create 00:17:25, use 00:01:54 timeout:86400000, left 23:58:05, Map-Id(In): 1,
    flags:
none, use_count: 0, entry-id: 32, lc_entries: 0
--- 209.165.200.227  192.168.11.10 ---          ---
    create 00:17:22, use 00:01:51 timeout:86400000, left 23:58:08, Map-Id(In): 1,
    flags:
none, use_count: 0, entry-id: 34, lc_entries: 0
R2#
```

| Command                                                                                                                                                                                       | Description                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <code>clear ip nat translation *</code>                                                                                                                                                       | Clears all dynamic address translation entries from the NAT translation table.                                    |
| <code>clear ip nat translation inside <i>global-ip</i> <i>local-ip</i> [<i>outside local-ip global-ip</i>]</code>                                                                             | Clear a simple dynamic translation entry containing an inside translation or both inside and outside translation. |
| <code>clear ip nat translation protocol <i>inside</i> <i>global-ip</i> <i>global-port</i> <i>local-ip</i> <i>local-port</i> [<i>outside local-ip local port global-ip global-port</i>]</code> | Clears an extended dynamic translation entry.                                                                     |

# Configure Dynamic NAT

## Verifying Dynamic NAT (Cont.)



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```
R2# clear ip nat statistics
```

PC1 and PC2 establish sessions with the server

```
R2# show ip nat statistics
```

```
Total active translations: 2 (0 static, 2 dynamic; 0 extended)
```

```
Peak translations: 6, occurred 00:27:07 ago
```

```
Outside interfaces:
```

```
Serial0/0/1
```

```
Inside interfaces:
```

```
Serial0/1/0
```

```
Hits: 24 Misses: 0
```

```
CEF Translated packets: 24, CEF Punted packets: 0
```

```
Expired translations: 4
```

```
Dynamic mappings:
```

```
-- Inside Source
```

```
[Id: 1] access-list 1 pool NAT-POOL1 refcount 2
```

```
pool NAT-POOL1: netmask 255.255.255.224
```

```
start 209.165.200.226 end 209.165.200.240
```

```
type generic, total addresses 15, allocated 2 (13%), misses 0
```

```
Total doors: 0
```

```
Appl doors: 0
```

```
Normal doors: 0
```

```
Queued Packets: 0
```

```
R2#
```

# Configure PAT

## Configuring PAT: Address Pool



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The pool contains the public addresses.

The ACL defines which private IP addresses gets translated.

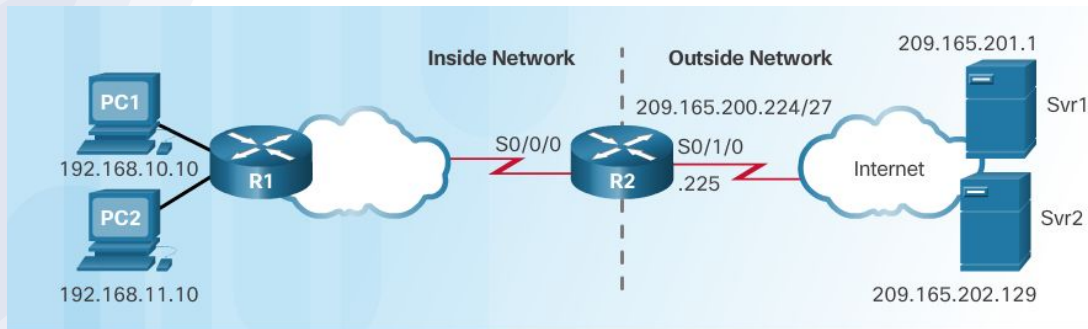
The **ip nat inside source list** *acl# pool name overload* command ties Step 1 with Step 2.

|        |                                                                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1 | Define a pool of global addresses to be used for overload translation.<br><br><code>ip nat pool name start-ip end-ip {netmask netmask prefix-length prefix-length}</code>       |
| Step 2 | Define a standard access list permitting the addresses that should be translated.<br><br><code>access-list access-list-number permit source [source-wildcard]</code>            |
| Step 3 | Establish overload translation, specifying the access list and pool defined in prior steps.<br><br><code>ip nat inside source list access-list-number pool name overload</code> |
| Step 4 | Identify the inside interface.<br><br><code>interface type number<br/>ip nat inside</code>                                                                                      |
| Step 5 | Identify the outside interface.<br><br><code>interface type number<br/>ip nat outside</code>                                                                                    |

The **overload** command is what allows the router to track port numbers (and do PAT instead of dynamic NAT).

# Configure PAT

## Configuring PAT: Address Pool (Cont.)



**Define a pool of public IPv4 addresses under the pool name NAT-POOL2.**

```
R2(config)# ip nat pool NAT-POOL2 209.165.200.226  
209.165.200.240 netmask 255.255.255.224
```

**Define which addresses are eligible to be translated.**

```
R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```

**Bind NAT-POOL2 with ACL 1.**

```
R2(config)# ip nat inside source list 1 pool NAT-POOL2  
overload
```

**Identify interface serial 0/0/0 as an inside NAT interface.**

```
R2(config)# interface Serial0/0/0  
R2(config-if)# ip nat inside
```

**Identify interface serial 0/1/0 as the outside NAT interface.**

```
R2(config)# interface Serial0/1/0  
R2(config-if)# ip nat outside
```

# Configure PAT

## Configuring PAT: Single Address



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- When a public address is assigned to the external interface on the border router, that public address can be used for PAT and translate internal private IP addresses to the public IP address.

Still need an ACL to define which private IP addresses gets translated.

Instead of associating an ACL with a pool, the ACL is associated with an interface that has a public IP address assigned.

### Step 1

Define a standard access list permitting the addresses that should be translated.

```
access-list access-list-number permit source [source-wildcard]
```

### Step 2

Establish dynamic source translation, specifying the ACL, exit interface and overload options.

```
ip nat inside source list access-list-number interface type number overload
```

### Step 3

Identify the inside interface.

```
interface type number  
ip nat inside
```

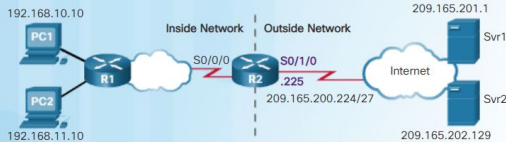
### Step 4

Identify the outside interface.

```
interface type number  
ip nat outside
```

The **overload** command is always needed for PAT.

PAT with Single Address



NAT Table

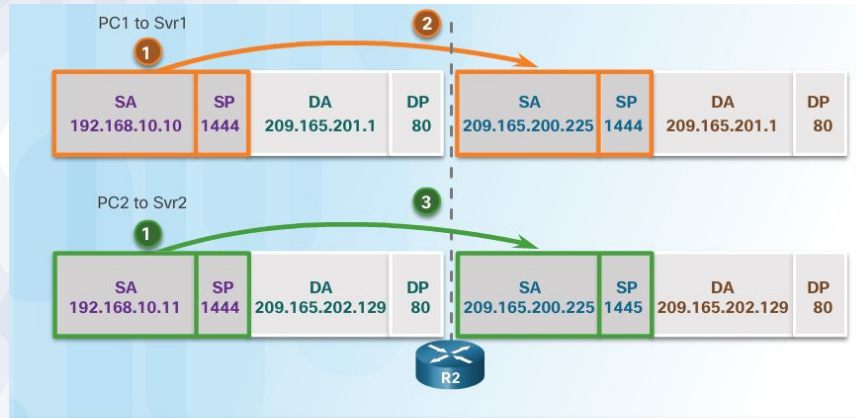
| Inside Global Address | Inside Local Address | Outside Global Address | Outside Local Address |
|-----------------------|----------------------|------------------------|-----------------------|
| 209.165.200.225:1444  | 192.168.10.10:1444   | 209.165.201.1:80       | 209.165.201.1:80      |
| 209.165.200.225:1445  | 192.168.11.10:1444   | 209.165.202.129:80     | 209.165.202.129:80    |

# Configure PAT

## Analyzing PAT



1. PC1 and PC2 open a web browser for a connection to a web server.
2. R2 receives the packets on the inside interface and checks if translation should be performed (via an ACL). R2 assigns the IP address of the outside interface, adds a port number, and creates a NAT table entry for both packets.
3. R2 replaces the inside local source address on each packet with the translated inside global address.



NAT Table

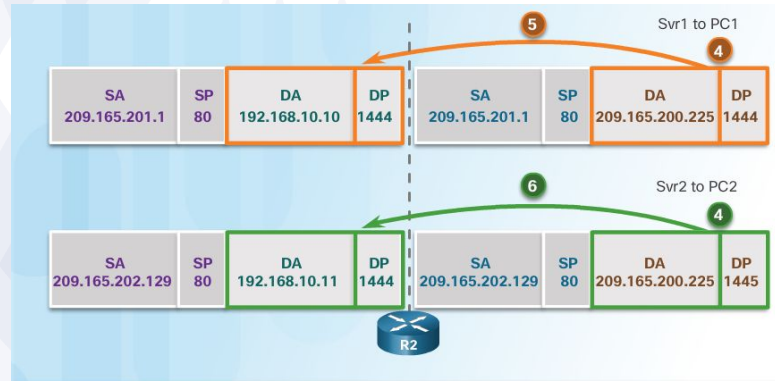
| Inside Local Address | Inside Global Address | Outside Global Address | Outside Local Address |
|----------------------|-----------------------|------------------------|-----------------------|
| 192.168.10.10:1444   | 209.165.200.225:1444  | 209.165.201.1:80       | 209.165.201.1:80      |
| 192.168.10.11:1444   | 209.165.200.225:1445  | 209.165.202.129:80     | 209.165.202.129:80    |

## Configure PAT

### Analyzing PAT (Cont.)



- Each server responds to PC1 and PC2 using the destination address of the public address assigned to the external interface on the border router.
- R2 looks up the received packet and forwards to PC1 because that is the private IP address found in the NAT table for the destination address and port number.
- R2 looks up the received packet and forwards to PC2 because that is the private IP address found in the NAT table for the destination address and port number.



NAT Table

| Inside Local Address | Inside Global Address | Outside Global Address | Outside Local Address |
|----------------------|-----------------------|------------------------|-----------------------|
| 192.168.10.10:1444   | 209.165.200.225:1444  | 209.165.201.1:80       | 209.165.201.1:80      |
| 192.168.10.11:1444   | 209.165.200.225:1445  | 209.165.202.129:80     | 209.165.202.129:80    |

# Configure PAT Verifying



```
R2# show ip nat translations
Pro Inside global          Inside local          Outside local          Outside global
tcp 209.165.200.226:51839  192.168.10.10:51839  209.165.201.1:80      209.165.201.1:80
tcp 209.165.200.226:42558  192.168.11.10:42558  209.165.202.129:80    209.165.202.129:80
R2#
```

```
R2# clear ip nat statistics

R2# show ip nat statistics
Total active translations: 2 (0 static, 2 dynamic; 2 extended)
Peak translations: 2, occurred 00:00:05 ago
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/1/0
Hits: 4 Misses: 0
CEF Translated packets: 4, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 3] access-list 1 pool NAT-POOL2 refcount 2
pool NAT-POOL2: netmask 255.255.255.224
  start 209.165.200.226 end 209.165.200.240
  type generic, total addresses 15, allocated 1 (6%), misses 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R2#
```

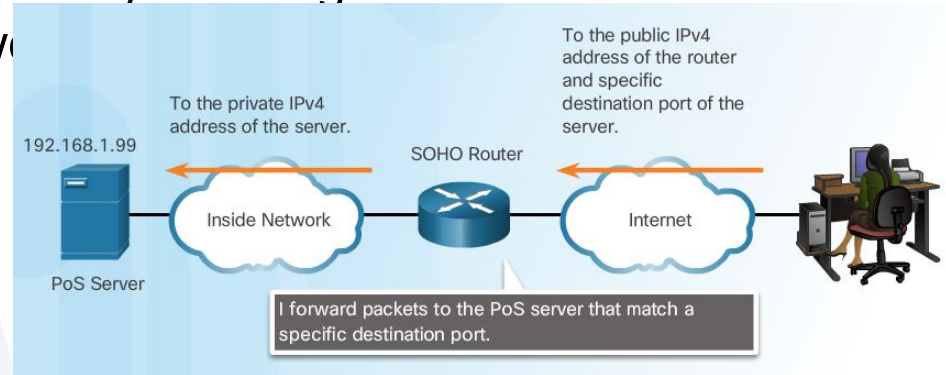
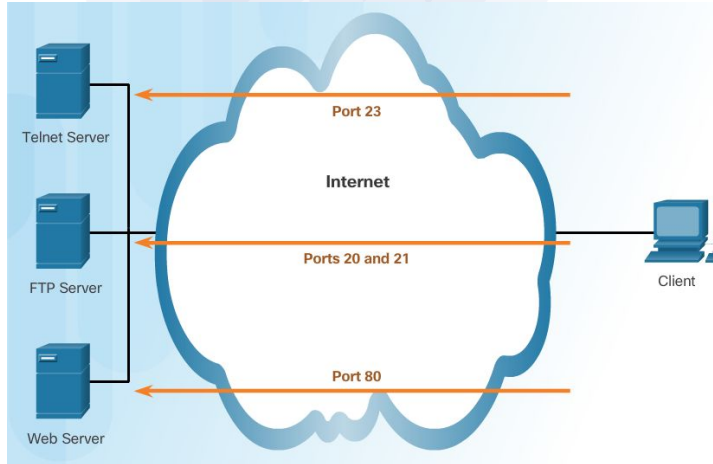
# Configure Port Forwarding

## Port Forwarding



- Port forwarding allows an external device to reach a device on a specific port number and the device is located on an internal (private) network.
  - Required for some peer-to-peer file-sharing programs and operations such as web serving and outgoing FTP

NAT only allowing translations for traffic  
etwo

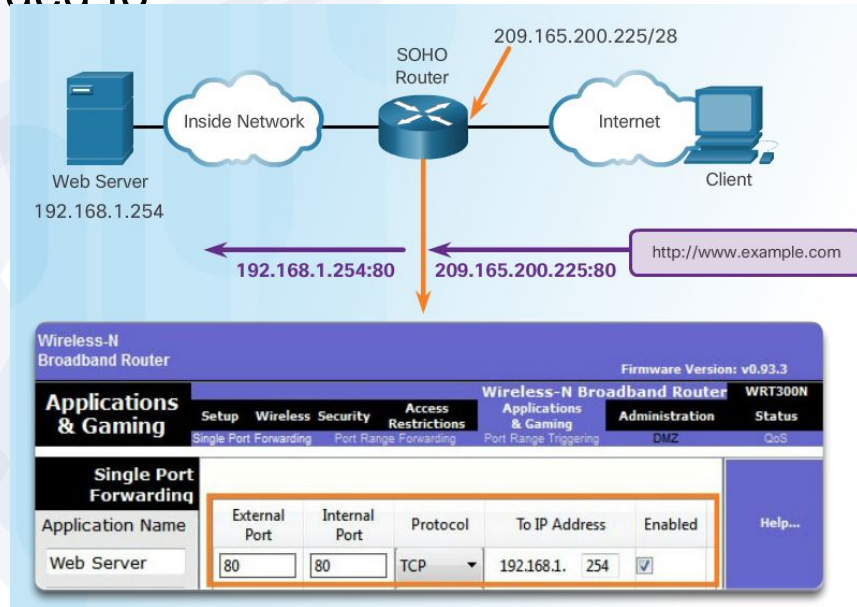


# Configure Port Forwarding Wireless Router Example



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- Port forwarding can be enabled for specific applications
  - Must specify the inside local address that requests should be forwarded to

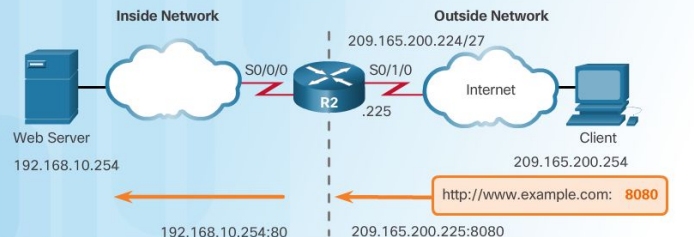


# Configure Port Forwarding

## Configuring Port Forwarding with IOS



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Establishes static translation between an inside local address and local port and an inside global address and global port.

```
R2(config)# ip nat inside source static tcp 192.168.10.254 80  
209.165.200.225 8080
```

Identifies interface serial 0/0/0 as an inside NAT interface.

```
R2(config)# interface Serial0/0/0  
R2(config-if)# ip nat inside
```

Identifies interface serial 0/1/0 as the outside NAT interface.

```
R2(config)# interface Serial0/1/0  
R2(config-if)# ip nat outside
```

```
ip nat inside source {static {tcp | udp local-ip local-port  
global-ip global-port} [extendable]}
```

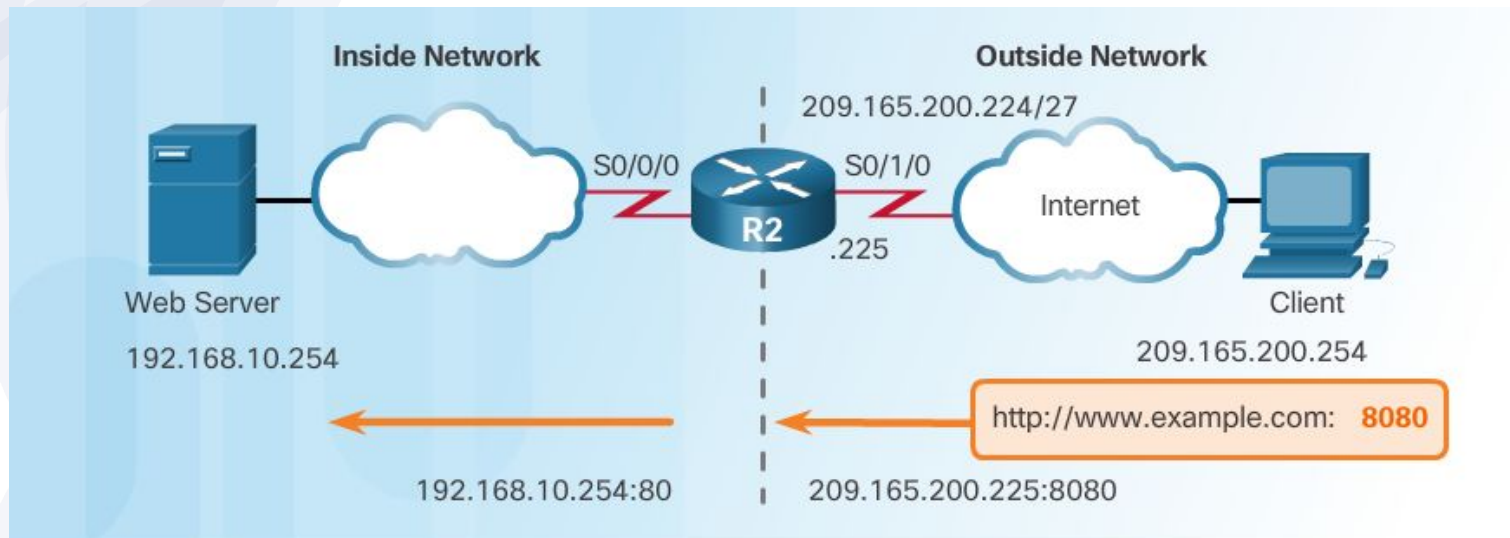
| Parameter                            | Description                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>tcp</code> or <code>udp</code> | Indicates if this is a TCP or UDP port number.                                                                                                                                                                                                                                                                                               |
| <code>local-ip</code>                | This is the IPv4 address assigned to the host on the inside network, typically from RFC 1918 private address space.                                                                                                                                                                                                                          |
| <code>local-port</code>              | Sets the local TCP/UDP port in a range from 1 – 65,535. This is the port number the server is listening on.                                                                                                                                                                                                                                  |
| <code>global-ip</code>               | This is the IPv4 globally unique IP address of an inside host. This is the IP address the outside clients will use to reach the internal server.                                                                                                                                                                                             |
| <code>global-port</code>             | Sets the global TCP/UDP port in a range from 1 – 65,535. This is the port number the outside client will use to reach the internal server.                                                                                                                                                                                                   |
| <code>extendable</code>              | The <code>extendable</code> option is applied automatically. The <code>extendable</code> keyword allows the user to configure several ambiguous static translations, where ambiguous translations are translations with the same local or global address. It allows the router to extend the translation to more than one port if necessary. |

# Configure Port Forwarding

## Configuring Port Forwarding with IOS (Cont.)



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```
R2# show ip nat translations
```

| Pro | Inside global        | Inside local      | Outside local         | Outside global        |
|-----|----------------------|-------------------|-----------------------|-----------------------|
| tcp | 209.165.200.225:8080 | 192.168.10.254:80 | 209.165.200.254:46088 | 209.165.200.254:46088 |
| tcp | 209.165.200.225:8080 | 192.168.10.254:80 | ---                   | ---                   |

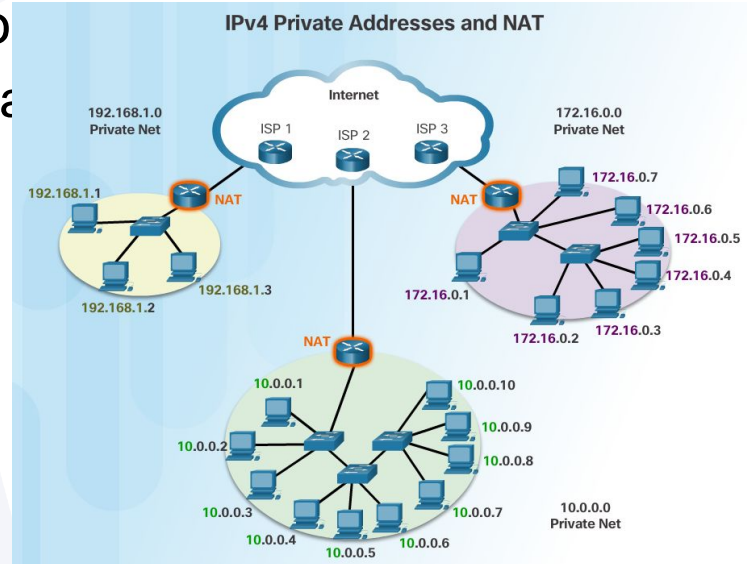
```
R2#
```

# NAT and IPv6

## NAT for IPv6?

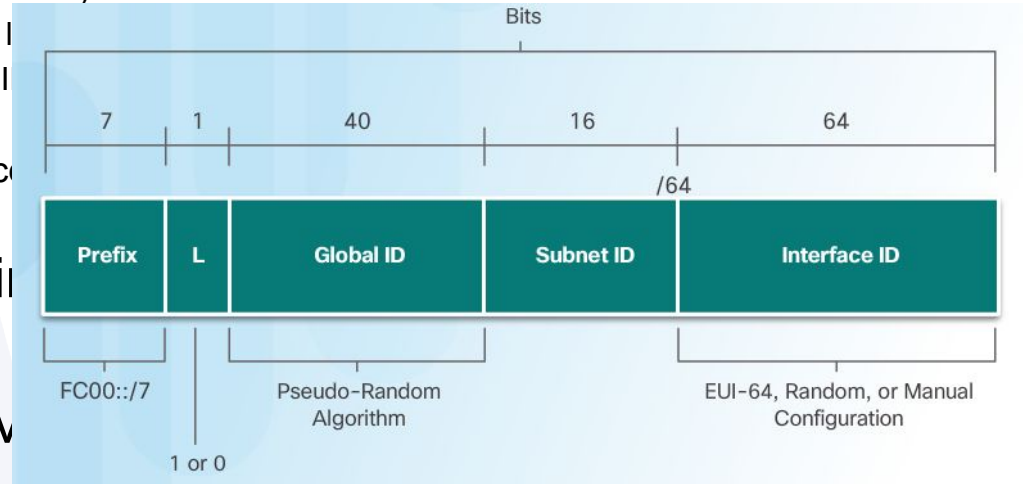


- IPv6 was developed with the intention of making NAT for IPv4 unnecessary
- IPv6 does have its own for  
– IPv6 has its own private address



- IPv6 unique local addresses (ULAs) are similar to IPv4 private addresses
  - ULAs are to provide IPv6 address space for communications within a local site.
  - First 64 bits of a ULA
    - Prefix of FC00::/7 (FC00 to FDFF)
    - Next bit is a 1 if the prefix is L
    - Next 40 bits define a global I
    - Next 16 bits is a subnet ID
  - Last 64 bits of a ULA is the interface or host portion of the address

- Allows sites to be combined without address conflicts
- Allows internal connectivity
- Not routable on the Internet

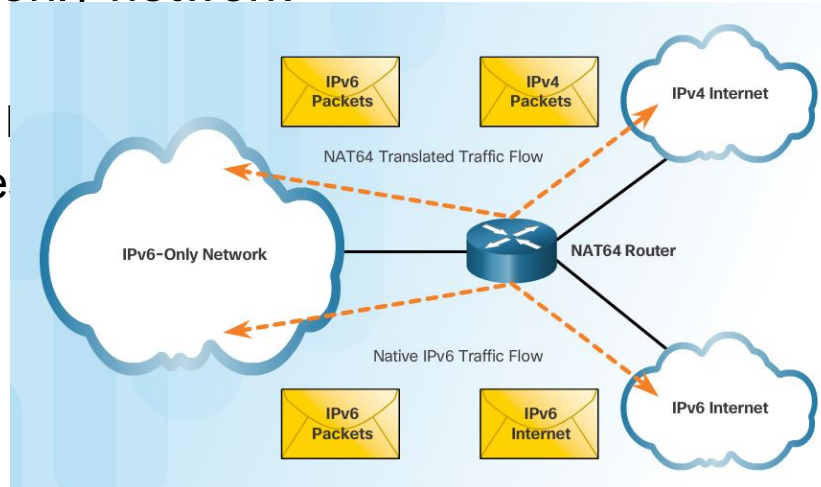


# NAT and IPv6

## NAT for IPv6



- Provide access between IPv6-only and IPv4-only networks (not translating private address to public addresses as NAT for IPv4 was)
- Techniques available
  - Dual-stack – both devices run protocols for both IPv4 and IPv6
  - Tunneling – Encapsulate the IPv6 packet inside an IPv4 packet for transmission over an IPv4-only network
  - NAT for IPv6 (translation)
    - Should not be used as a
    - The older Network Address Translation (NAT-PT)
    - NAT64





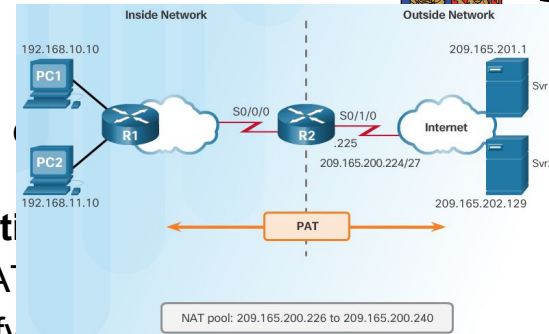
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# Troubleshoot NAT

# NAT Troubleshooting Commands

## The show ip nat Commands

1. Determine what NAT is supposed to achieve and (if applicable) what the problem with the configuration.
2. Verify translations using the **show ip nat translations** command.
3. Use the **clear** and **debug** commands to verify NAT operation.
4. Review what is happening to the packet and verify the results.



```
R2# clear ip nat statistics
R2# clear ip nat translation *
R2#
<output omitted>

R2# show ip nat statistics
Total active translations: 1 (0 static, 1 dynamic; 1 extended)
Peak translations: 1, occurred 00:00:09 ago
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/0/0
Hits: 31 Misses: 0
CEF Translated packets: 31, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 5] access-list 1 pool NAT-POOL2 refcount 1
  pool NAT-POOL2: netmask 255.255.255.224
  start 209.165.200.226 end 209.165.200.240
  type generic, total addresses 15, allocated 1 (6%), misses 0
<output omitted>

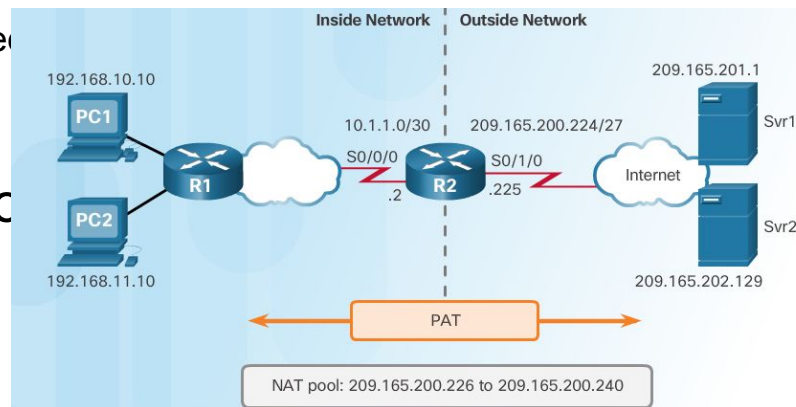
R2# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 209.165.200.226:19005 192.168.10.10:19005 209.165.201.1:23 209.165.201.1:23
```

# NAT Troubleshooting Commands

## The debug ip nat Command

- Common commands
  - debug ip nat**
  - debug ip nat detailed**
- Output symbols and values
  - \* - The translation is occurring in the fast
  - s= - Source IPv4 address
  - a.b.c.d--->w.x.y.z – Source a.b.c.d is translated to w.x.y.z
  - d= - Destination IPv4 address
  - [xxxx] - IPv4 identification number
- Check the ACL to ensure the correct addresses are designated.

```
R2# debug ip nat
IP NAT debugging is on
R2#
*Feb 15 20:01:311.670: NAT*: s=192.168.10.10->209.165.200.226, d=209.165.201.1 [2817]
*Feb 15 20:01:311.682: NAT*: s=209.165.201.1, d=209.165.200.226->192.168.10.10 [4180]
*Feb 15 20:01:311.698: NAT*: s=192.168.10.10->209.165.200.226, d=209.165.201.1 [2818]
*Feb 15 20:01:311.702: NAT*: s=192.168.10.10->209.165.200.226, d=209.165.201.1 [2819]
*Feb 15 20:01:311.710: NAT*: s=192.168.10.10->209.165.200.226, d=209.165.201.1 [2820]
*Feb 15 20:01:311.710: NAT*: s=209.165.201.1, d=209.165.200.226->192.168.10.10 [4181]
*Feb 15 20:01:311.722: NAT*: s=209.165.201.1, d=209.165.200.226->192.168.10.10 [4182]
*Feb 15 20:01:311.726: NAT*: s=192.168.10.10->209.165.200.226, d=209.165.201.1 [2821]
*Feb 15 20:01:311.730: NAT*: s=209.165.201.1, d=209.165.200.226->192.168.10.10 [4183]
*Feb 15 20:01:311.734: NAT*: s=192.168.10.10->209.165.200.226, d=209.165.201.1 [2822]
*Feb 15 20:01:311.734: NAT*: s=209.165.201.1, d=209.165.200.226->192.168.10.10 [4184]
<output omitted>
```



```
R2# show access-lists
Standard IP access list 1
10 permit 192.168.0.0, wildcard bits 0.0.255.255 (29 matches)
```

# NAT Troubleshooting Commands

## NAT Troubleshooting Scenarios

- Internal hosts cannot contact external servers.

```
R2# show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  Serial0/0/0
Inside interfaces:
  Serial0/1/0
Hits: 0 Misses: 0
<output omitted>
```

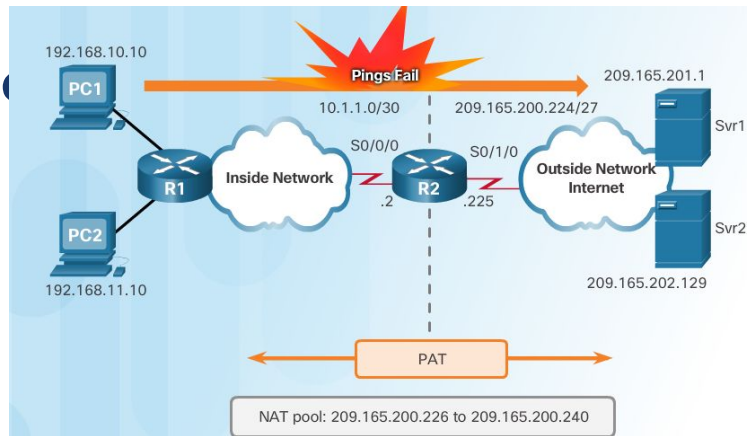
2. Outside and inside  
interfaces are reversed

```
R2(config)# interface serial 0/0/0
R2(config-if)# no ip nat outside
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface serial 0/0/1
R2(config-if)# no ip nat inside
R2(config-if)# ip nat outside
```

```
R2# show ip nat statistics
Total active translations: 1 (0 static, 1 dynamic; 1 extended)
Peak translations: 1, occurred 00:37:58 ago
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/1/0
Hits: 20 Misses: 0
CEF Translated packets: 20, CEF Punted packets: 0
Expired translations: 1
Dynamic mappings:
-- Inside Source
[Id: 5] access-list 1 pool NAT-POOL2 refcount 1
  pool NAT-POOL2: netmask 255.255.255.224
  start 209.165.200.226 end 209.165.200.240
  type generic, total addresses 15, allocated 1 (6%), misses 0
<output omitted>

R2# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 209.165.200.226:38 192.168.10.10:38  209.165.201.1:38   209.165.201.1:38
R2#
```

Translations  
working!



```
R2# show ip nat translations
R2#
```

1. No translations in NAT table

```
R2# show access-lists
Standard IP access list 1
  10 permit 192.168.0.0, wildcard bits 0.0.0.255
R2#
```

3. Incorrect ACL

```
R2(config)# no access-list 1
R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```